Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which one of the following is the wrong assumption of kinetic theory of gases?

(A) Momentum and energy always remain conserved.

(B) Pressure is the result of elastic collision of molecules with the container's wall.

(C) Molecules are separated by great distances compared to their sizes.

(D) All the molecules move in straight line between collision and with same velocity.

Q.2

A He atom at $300\ \mathrm{K}$ is released from the surface of the earth to travel upwards. Assuming that it undergoes no collision with other molecules, how high will it be before coming to the rest?

(A) $9.53\ \mathrm{m}$

(B) $95.3\ \mathrm{m}$

(C) $953\ \mathrm{m}$

(D) $9.53 \times 10^4\ \mathrm{m}$

Q.3

Which of the following is not an assumption of the kinetic theory of gases?

(A) Gas particles have negligible volume.

(B) A gas consists of many identical particles which are in continual motion.

(C) At high pressure, gas particles are difficult to compress.

(D) Collisions of gas particles are perfectly elastic.

Q.4

The temperature at which oxygen molecules have the same root mean square speed as helium atoms have at $300\ \mathrm{K}$ is:
Atomic masses: $\mathrm{He} = 4\mathrm{u}, \mathrm{O} = 16\mathrm{u}$)

(A) $300\ \mathrm{K}$

(B) $600\ \mathrm{K}$

(C) $1200\ \mathrm{K}$

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(D) $2400\ \mathrm{K}$

Q.5

By how many folds the temperature of a gas would increase when the root mean square velocity of the gas molecules in a container of fixed volume is increased from $5 \times 10^4\ \mathrm{cm/s}$ to $10 \times 10^4\ \mathrm{cm/s}$?

(A) Two

(B) Three

(C) Six

(D) Four

Q.6

α, v and u represent most probable velocity, average velocity and root mean square velocity respectively of a gas at a particular temperature. The correct order among the following is

(A) $u > v > \alpha$

(B) $v > u > \alpha$

(C) $\alpha > u > v$

(D) $u > \alpha > v$

Q.7

The most probable speed of $8\ \mathrm{g}$ of $\mathrm{H_2}$ is $200\ \mathrm{ms^{-1}}$. Average kinetic energy (neglect rotational and vibrational energy) of $\mathrm{H_2}$ gas is :

(A) $480\ \mathrm{J}$

(B) $240\ \mathrm{J}$

(C) $120\ \mathrm{J}$

(D) none of these

Q.8

Helium atom is two times heavier than a hydrogen molecule at. $298\ \mathrm{K}$. The average KE of helium is

(A) 2 times of $\mathrm{H_2}$ molecule

(B) same as that of $\mathrm{H_2}$ molecule

(C) 4 times that of hydrogen molecule

(D) $\frac{1}{2}$ that of $\mathrm{H_2}$ molecule

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Q.9

The molecular velocities of two gases at the same temperature are u_1 and u_2 and their masses are m_1 and m_2 respectively. Which of the following expressions are correct?

(A) $\frac{m_1}{u_1^2} = \frac{m_2}{u_2^2}$

(B) $m_1 u_1 = m_2 u_2$

(C) $\frac{m_1}{u_1} = \frac{m_2}{u_2}$

(D) $m_1 u_1^2 = m_2 u_2^2$

Q.10

At what temperature will the r.m.s. velocity of oxygen be one and half times of its value at N.T.P.?

(A) $614.25\ \text{K}$

(B) $525\ \text{K}$

(C) $519.25\ \text{K}$

(D) $114.25\ \text{K}$

Q.11

The average speed of an ideal gas molecule at 27°C is $0.3\ \text{msec}^{-1}$. Calculate average speed at 927°C.

(A) $0.6\ \text{msec}^{-1}$

(B) $1.0\ \text{msec}^{-1}$

(C) $1.4\ \text{msec}^{-1}$

(D) $2.8\ \text{msec}^{-1}$

Q.12

What is kinetic energy of 1g of O_2 at 47°C

(A) $1.24 \times 10^2\ \text{J}$

(B) $2.24 \times 10^2\ \text{J}$

(C) $1.24 \times 10^3\ \text{J}$

(D) $3.24 \times 10^2\ \text{J}$

Q.13

By what ratio the average velocity of the molecule in gas change when the temperature is raised from 50 to 200°C

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(A) $1.21 : 1$

(B) $1.46 : 1$

(C) $1.14 : 1$

(D) $4 : 1$

Q.14

The rms speed of hydrogen is $\sqrt{7}$ times the ms speed of nitrogen, if T is the temperature of the gas then

(A) $T_{H_2} = T_{N_2}$

(B) $T_{H_2} > T_{N_2}$

(C) $T_{A_2} < T_{N_2}$

(D) $T_{H_2} > \sqrt{7} T_{N_2}$

Q.15

The temperature of the gas is raised from $27^\circ \mathrm{C}$ to $927^\circ \mathrm{C}$, the root mean square velocity is

(A) $\sqrt{927/27}$ time the earlier value

(B) same as before

(C) halved

(D) doubled

Q.16

The ratio between the root mean square speed of $\mathrm{H_2}$ at $50\ \mathrm{K}$ and that of $\mathrm{O_2}$ at $800\ \mathrm{K}$ is,

(A) 4

(B) 2

(C) 1

(D) $1/4$

Q.17

As the temperature is raised from $20^\circ \mathrm{C}$ to $40^\circ \mathrm{C}$, the average kinetic energy of neon atoms changes by a factor of which of the following ?

(A) $313/293$

(B) $\sqrt{(313/293)}$

(C) $1/2$

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(D) 2

Q.18

Which one of the following statements is NOT true about the effect of an increase in temperature on the distribution of molecular speeds in a gas?

(A) The area under the distribution curve remains the same as under the lower temperature.

(B) The distribution becomes broader.

(C) The fraction of the molecules with the most probable speed increases.

(D) The most probable speed increases.

Q.19

For gaseous state, if most probable speed is denoted by C^*, average speed by $\overline{C}$ and mean square speed by C, then for large number of molecules the ratios of these speeds are :

(A) $C^* : \overline{C} : C = 1.225 : 1.128 : 1$

(B) $C^* : \overline{C} : C = 1.128 : 1.225 : 1$

(C) $C^* : \overline{C} : C = 1 : 1.128 : 1.225$

(D) $C^* : \bar{C} : C = 1 : 1.225 : 1.128$

Q.20

Let the most probable velocity of hydrogen molecules at a temperature of $t^\circ C$ be V_0. When the temperature is raised to $(2t + 273)^\circ C$ the new rms velocity is (suppose all the molecules dissociate into atoms at latter temperature)

(A) $2\sqrt{3}\, V_0$

(B) $\sqrt{6}\, V_0$

(C) $\sqrt{3\left(2 + \frac{273}{t}\right)} V_0$

(D) $\sqrt{\frac{2}{3}}\, V_0$

Answers & Solutions

Q.1 Answer:

All the molecules move in straight line between collision and with same velocity.

Solution:

Molecules move very fast in all directions in a straight line by colliding with each other but with different velocity.

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Q.2 Answer:

9.53×10^4 m

Solution:

$\frac{3}{2}k_B T = mgh$

$h = \frac{3}{2} \times \frac{1.38 \times 10^{-23} \times 300}{9.81 \times 4 \times 1.66 \times 10^{-27}} = 9.53 \times 10^4$ m

Q.3 Answer:

At high pressure, gas particles are difficult to compress.

Solution:

At high pressure real gas particles are easily compressed.

Q.4 Answer:

2400 K

Solution:

$V_{rms} = \sqrt{\frac{3RT}{M}}$

$V_{ms(O_2)} = V_{rms(He)}$

$\sqrt{\frac{3RT_{O_2}}{M_{O_2}}} = \sqrt{\frac{3RT_{He}}{M_{He}}}$

or $\frac{T_{O_2}}{M_{O_2}} = \frac{T_{He}}{M_{He}}$

$\therefore \quad T_{O_2} = \frac{300 \times 32}{4} = 2400$ K

Q.5 Answer:

Four

Solution:

r.m.s. velocity $V_{rms} = \sqrt{\frac{3RT}{M}}$

i.e., $\frac{V_1}{V_2} = \sqrt{\frac{T_1}{T_2}}$

$\frac{5 \times 10^4}{10 \times 10^4} = \frac{1}{2} = \sqrt{\frac{T_1}{T_2}}$

$\therefore \quad T_2 = 4\, T_1$

Q.6 Answer:

$u > v > \alpha$

Solution:

$u : v : \alpha = \sqrt{\frac{3RT}{M}} : \sqrt{\frac{8RT}{\pi M}} : \sqrt{\frac{2RT}{M}}$

$= \sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2}$

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Q.7 Answer:

$240\ \mathrm{J}$

Solution:

$200 = \sqrt{\frac{2\mathrm{RT}}{2\times10^{-3}}}; \mathrm{RT} = 40$

Average K.E. $= \frac{3}{2}\mathrm{nRT} = \frac{3}{2} \times \frac{8}{2} \times 40 = 240\ \mathrm{J}$

Q.8 Answer:

same as that of $\mathrm{H_2}$ molecule

Solution:

$\mathrm{KE} = \frac{3}{2}\frac{\mathrm{n}}{\mathrm{N_o}}\mathrm{RT}$. **Average** $\overline{\mathrm{KE}}$ **is the same for all gaseous molecules**

Q.9 Answer:

$\mathrm{m_1u_1^2} = \mathrm{m_2u_2^2}$

Solution:

$U_1 \propto \sqrt{\frac{1}{m_1}}$ **and** $U_2 \propto \sqrt{\frac{1}{m_2}}$

$\therefore \frac{U_1^2}{U_2^2} = \frac{m_2}{m_1} \quad \therefore m_1U_1^2 = m_2U_2^2$

Q.10 Answer:

$614.25\ \mathrm{K}$

Solution:

$\frac{1}{2}\mathrm{mc^2} = \frac{3}{2}\mathrm{KT}$

Suppose the temperature required is $\mathrm{T'}$ **when the velocity will be** $\frac{3}{2}\mathrm{C}$

$\frac{3/2C}{C} = \sqrt{\frac{T'}{T}}$ **or,** $T' = \frac{9}{4} \times 273 = 614.25\ \mathrm{K}$

Q.11 Answer:

$0.6\ \mathrm{msec^{-1}}$

Solution:

$\mathbf{u_{av}} = \sqrt{\frac{8\mathrm{RT}}{\pi\mathrm{M}}}$

Given $u_{\mathrm{av}} = 0.3\ \mathrm{msec^{-1}}$ **at 300K** $\quad u_1 = 0.3 = \sqrt{\frac{8R\times300}{\pi M}}$

at $\mathrm{T} = 273 + 927 = 1200\ \mathrm{K}$

$\mathbf{u_2} = \sqrt{\frac{8R\times1200}{\pi M}}$

$\frac{\mathbf{u_2}}{0.3} = \sqrt{\frac{1200}{300}}\mathbf{u_2} = 0.6\ \mathrm{msec^{-1}}$

Q.12 Answer:

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1.24×10^2 J

Solution:

$K, E = \frac{3}{2}nRT = \frac{3}{2} \times \frac{1}{32} \times 8.314 \times 320^\circ \mathrm{K}$
$= 1.24 \times 10^2$ J

Q.13 Answer:

$1.21 : 1$

Solution:

$\mathrm{T}_1 = 200 + 273 = 473; \mathrm{T}_2 = 50 + 273 = 323$ K
Hence, $\frac{(\mathrm{V_{gv}})_1}{(\mathrm{V_{av}})_2} = \sqrt{\frac{\mathrm{T}_1}{\mathrm{T}_2}} = \sqrt{\frac{473}{323}} = 1.210 = 1.21 : 1$

Q.14 Answer:

$T_{A_2} < T_{N_2}$

Solution:

$\mathrm{U_{ms}}\,(\mathrm{H_2}) = \sqrt{7}\mathrm{U_{ms}}\,(\mathrm{N_2})$
$\sqrt{\frac{2rT_{\mathrm{H_2}}}{2}} = \sqrt{7} \times \sqrt{\frac{3RT_{N_2}}{28}}$
$\therefore \mathrm{T_{N_2}} = 2\ \mathrm{T_{H_2}}$ **or**
$\mathbf{T_{N_2}} > \mathbf{T_{H_2}}$

Q.15 Answer:

doubled

Solution:

$u \alpha \sqrt{T}$ **or** $u_1/u_2 = \sqrt{T_1/T_2}$
$= \sqrt{\frac{27+273}{927+273}} = \sqrt{\frac{300}{1200}} = \frac{1}{2}$
$u_2 = 2u_1$

Q.16 Answer:

1

Solution:

The expression of root mean square speed is $u_{rms} = \sqrt{\frac{3RT}{M}}$

Hence,

$\frac{\mathrm{u_{rms}(H_2)}}{\mathrm{u_{rms}(O_2)}} = \left[\frac{3\mathrm{R}(50\ \mathrm{K})/\left(2\ \mathrm{g\ mol^{-1}}\right)}{3\mathrm{R}(800\ \mathrm{K})/(32\ \mathrm{g\ mol^{-1}})}\right]^{1/2} = 1$

Q.17 Answer:

$313/293$

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Solution:

$$\frac{\text{K.E of neon at } 40^{\circ}\text{C}}{\text{K.E of neon at } 20^{\circ}\text{C}} = \frac{\frac{3}{2}\text{ K}\times 313}{\frac{3}{2}\text{ K}\times 293} = \frac{313}{293}$$

Q.18 Answer:

The fraction of the molecules with the most probable speed increases.

Solution:

As temperature rises the most probable speed increases and the fraction of molecules possessing most probable speed decreases.

Q.19 Answer:

$\mathrm{C}^{*} : \overline{\mathrm{C}} : \mathrm{C} = 1 : 1.128 : 1.225$

Solution:

Most probable speed $\left(\mathrm{C}^{*}\right) = \sqrt{\frac{2\mathrm{RT}}{\mathrm{M}}}$

Average Speed $\left(\overline{\mathrm{C}}\right) = \sqrt{\frac{8\mathrm{RT}}{\pi\mathrm{M}}}$

Root mean square velocity $\left(\mathrm{C}\right) = \sqrt{\frac{3\mathrm{RT}}{\mathrm{M}}}$

$$\mathrm{C}^{*} : \overline{\mathrm{C}} : \mathrm{C} = \sqrt{\frac{2\mathrm{RT}}{\mathrm{M}}} : \sqrt{\frac{8\mathrm{RT}}{\pi\mathrm{M}}} : \sqrt{\frac{3\mathrm{RT}}{\mathrm{M}}}$$

$$= 1 : \sqrt{\frac{4}{\pi}} : \sqrt{\frac{3}{2}} = 1 : 1.128 : 1.225$$

Q.20 Answer:

$\sqrt{6}\ \mathrm{V}_0$

Solution:

As we know that,

$$\mathrm{V}_{(\text{Most probable})} = \sqrt{\frac{2\mathrm{RT}}{\mathrm{M}}} \text{ ... (1)}$$

$$\text{At } \mathrm{T} = \mathrm{t}^{\circ}\mathrm{C} = (\mathrm{t} + 273)\mathrm{K}$$

$$\mathrm{M} = 2\text{ g}$$

$$\mathrm{V}_{(\mathrm{MP.})} = \mathrm{V}_0$$

$$\therefore \mathrm{V}_{(\mathrm{MP.})} = \sqrt{\frac{2\mathrm{R}(\mathrm{t}+273)}{2}} = \sqrt{\mathrm{R}(\mathrm{t}+273)}$$

$$\Rightarrow \mathrm{V}_0 = \sqrt{\mathrm{R}(\mathrm{t}+273)}$$

$$\text{At } \mathrm{T} = (2\mathrm{t} + 273)^{\circ}\mathrm{C} = 2(\mathrm{t}+273)\mathrm{K}$$

Since the molecule is dissociated into atom.

$$\mathrm{M} = 1\text{ g}$$

$$\therefore \mathrm{V}_{(\mathrm{MP})} = \sqrt{\frac{2\mathrm{R}\cdot 2(\mathrm{t}+273)}{1}} = 2\sqrt{\mathrm{R}(\mathrm{t}+273)} = 2\ \mathrm{V}_0$$

Now,

$$\mathrm{V}_{(\mathrm{rms.})} = \sqrt{\frac{3\mathrm{RT}}{\mathrm{M}}} \text{ ... (3)}$$

From equation (1) and (3), we get

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$V_{(rms.)} = \sqrt{\frac{3}{2}} \times V_{(MP)}$

$\therefore V_{(rms.)} = \sqrt{\frac{3}{2}} \times 2\, V_0 = \sqrt{6}\, V_0$

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